

HW05 Performance-Testing Report

1. Introduction

This report summarizes the HW05 performance-testing submission. It covers the Load, Stress, Spike, and Endurance evidence already present in the repository.

2. System Under Test

- SUT: EShop backend API
- Backend source: `../apps/backend/server.js`
- Database: SQLite in `../apps/backend/database.js`
- Base URL: `http://localhost:3000`

3. Test Environment

Hardware evidence is stored in `../evidence/hardware/hardware.png`.

Observed values in the screenshot:

Item	Value
Computer name	DESKTOP-0E72VR4
OS	Windows 11 Home Single Language 64-bit (10.0, Build 26200)
System manufacturer	ASUSTeK COMPUTER INC.
System model	Vivobook_ASUSLaptop X1505VA_A1505VA
BIOS	X1505VA.300
CPU	13th Gen Intel(R) Core(TM) i9-13900H (20 CPUs), ~2.6GHz
Installed RAM	16384MB RAM

4. Test Data

The JMeter workflow uses the existing CSV files in `../jmeter/data/`:

- `admin-users.csv`
- `users.csv`
- `products.csv`
- `checkout-data.csv`
- `workflow-data.csv`

The workflow covers these endpoint groups:

- authentication: `POST /api/login`
- transactional order creation: `POST /api/checkout`
- read-heavy order inspection: `GET /api/admin/orders`

- order administration and state update: [PUT /api/admin/orders/:id/status](#)

5. Performance Testing Methodology

The existing JMeter plans are:

- [../jmeter/plans/23127543_Load_20260813.jmx](#)
- [../jmeter/plans/23127543_Stress_20260813.jmx](#)
- [../jmeter/plans/23127543_Spike_20260813.jmx](#)

Observed scenario parameters in the JMX files:

Scenario	Threads	Ramp-up	Duration	Think time
Load	10	60 s	300 s	250 ms
Stress	30	90 s	600 s	150 ms
Spike	50	8 s	120 s	50 ms

6. Load Test

Evidence:

- Raw JTL: [../jmeter/results/load/load-20260817-142253.jtl](#)
- HTML report: [../jmeter/results/load/report-20260817-142253/](#)

Results:

Metric	Value
Samples	10,085
Success	10,085
Failures	0
Error rate	0.00%
Average response time	12.56 ms
Median	10 ms
P95-equivalent percentile	45.14 ms
Min	2 ms
Max	614 ms
Throughput	33.69 req/s

7. Stress Test

Evidence:

- Raw JTL: [../jmeter/results/stress/stress-20260817-143153.jtl](#)

- HTML report: [../jmeter/results/stress/report-20260817-143153/](#)

Results:

Metric	Value
Samples	26,866
Success	26,866
Failures	0
Error rate	0.00%
Average response time	464.21 ms
Median	512 ms
P95-equivalent percentile	1627.99 ms
Min	3 ms
Max	2887 ms
Throughput	44.77 req/s

8. Spike Test

Evidence:

- Raw JTL: [../jmeter/results/spike/spike-20260817-144417.jtl](#)
- HTML report: [../jmeter/results/spike/report-20260817-144417/](#)

Results:

Metric	Value
Samples	19,033
Success	19,033
Failures	0
Error rate	0.00%
Average response time	4324.65 ms
Median	4255 ms
P95-equivalent percentile	8419.64 ms
Min	3 ms
Max	13,834 ms
Throughput	21.08 req/s

9. Endurance Testing

The repository contains an endurance run in `../jmeter/results/endurance/endurance-20260818-083624.jtl` and its HTML report in `../jmeter/results/endurance/report-20260818-083624/`.

Measured endurance evidence from the current run:

- Duration: 900 seconds
- Threads: 100
- Ramp-up: 60 seconds
- Loops: -1
- Total samples: 19,033
- Total errors: 0
- Total throughput: 21.08 req/s
- Total average response time: 4324.65 ms
- Total median response time: 4255 ms
- Total p95-equivalent percentile: 8419.64 ms

10. Results and Performance Issues

- Load completed with zero sampler failures.
- Stress completed with zero sampler failures.
- Spike completed with zero sampler failures.
- Endurance completed with zero sampler failures.

Confirmed functional bugs from the raw JTL evidence: 0

Observed performance issues with significant latency inflation: 2 scenarios

- Stress
- Spike

11. AI-Analysis Critique

The critique of AI assistance is preserved in `../ai/AI_Critique.md`.

Key point from that critique:

- AI was helpful for explaining performance-testing concepts and troubleshooting, but its suggestions were not always reliable and had to be validated against the actual JMeter results and evidence.